

Some (potentially) helpful formulas:

Student #2

Thermodynamics:

Kinetic Theory of Gases

$$dE = dQ + dW$$

$$C_X = \left. \frac{dQ}{dT} \right|_X$$

$$dE = TdS + \sum_i J_i dx_i + \mu dN$$

$$F = E - TS$$

$$G = E - TS - Jx$$

$$\Phi = E - TS - \mu N$$

$$\frac{d\vec{q}_i}{dt} = \frac{\partial H}{\partial \vec{p}_i}$$

$$\frac{d\vec{p}_i}{dt} = -\frac{\partial H}{\partial \vec{q}_i}$$

$$\{A, B\} = \sum_{\alpha=1}^{3N} \left(\frac{\partial A}{\partial q_\alpha} \cdot \frac{\partial B}{\partial p_\alpha} - \frac{\partial A}{\partial p_\alpha} \cdot \frac{\partial B}{\partial q_\alpha} \right)$$

$$\frac{\partial \rho}{\partial t} = \{H, \rho\}$$

Probability/Mathematical Formulas:

Classical Statistical Mechanics

$$\tilde{p}(k) = \langle e^{-ikx} \rangle = \sum_{n=0}^{\infty} \frac{(-ik)^n}{n!} \langle x^n \rangle$$

$$\ln \tilde{p}(k) = \sum_{n=1}^{\infty} \frac{(-ik)^n}{n!} \langle x^n \rangle_c$$

$$\int_0^{\infty} dx x^n e^{-\alpha x} = \frac{n!}{\alpha^{n+1}}$$

$$\int_{-\infty}^{\infty} dx e^{-ikx - \frac{x^2}{2\sigma^2}} = \sqrt{2\pi\sigma^2} e^{-\frac{\pi^2 k^2}{2}}$$

$$\int_{-\infty}^{\infty} x^2 e^{-ax^2} = \frac{1}{2} \sqrt{\frac{\pi}{a^3}}$$

$$\lim_{x \rightarrow \infty} N! = N \ln N - N$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n!}$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

$$\sinh x = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$$

$$\cosh x = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

$$p_{(T,x,N)}(\mu_s) = \frac{e^{-\beta H(\mu_s)}}{Z(T,x,N)}$$

$$Z(T,x,N) = \sum_{\mu_s} e^{-\beta H(\mu_s)}$$

$$p_{(T,J,N)}(\mu_s) = \frac{e^{\beta \cdot J \cdot x - \beta H(\mu_s)}}{Z(T,J,N)}$$

$$Z(T,J,N) = \sum_{\mu_s} e^{-\beta(H(\mu_s) - Jx)}$$

$$p_{(T,\mu,x)}(\mu_s) = \frac{e^{\beta \cdot J \cdot x - \beta H(\mu_s)}}{Z(T,J,N)}$$

$$Q(T,\mu,x) = \sum_{\mu_s} e^{-\beta(H(\mu_s) - \mu N)}$$

$$S = k \ln \Omega$$

$$F = -k_B T \ln Z(T,x,N)$$

$$G = -k_B T \ln Z(T,J,N)$$

$$\Phi = -k_B T \ln Q(T,\mu,x)$$

$$\langle E \rangle = -\frac{\partial \ln Z}{\partial \beta}$$

$$\langle x \rangle = k_B T \frac{\partial \ln Z}{\partial J}$$

$$\langle N \rangle = \frac{\partial \ln Q}{\partial (\beta \mu)}$$

$$\lambda(T) = \frac{h}{\sqrt{2\pi m k_B T}}$$

Interacting Particles

$$\ln Z = \ln Z_0 + \frac{N(N-1)}{2V} \int d^3\vec{q} [\exp(-\beta V(\vec{q})) - 1] + O\left(\frac{N^3}{V^3}\right)$$

$$\frac{P}{k_B T} = \frac{N}{V} \left[1 + B_2(T) \frac{N}{V} + B_3(T) \frac{N^2}{V^2} + \dots \right]$$

$$f(\vec{q}) = \exp(-\beta V(\vec{q})) - 1$$

$$B_l(T) = -\frac{l-1}{l!} \bar{d}_l$$

$$(P - an^2)(V - Nb) = Nk_B T$$

$$Z(T, N, V) \approx \frac{(V - N\Omega/2)^2}{N! \lambda^{3N}} \exp \frac{\beta \mu N^2}{2V}$$

Quantum Statistical Mechanics

$$\eta = \begin{cases} +1 & \text{for bosons} \\ -1 & \text{for fermions} \end{cases}$$

$$\ln Q_\eta = -\eta \sum_{\vec{k}} \ln \left[1 - \eta \exp \left(\beta \mu - \beta \epsilon(\vec{k}) \right) \right]^{-1}$$

$$z = e^{\beta \mu}$$

$$N_\eta = \sum_{\vec{k}} \frac{1}{z^{-1} e^{\beta \epsilon(\vec{k})} - \eta}$$

$$E_\eta = \sum_{\vec{k}} \frac{\epsilon(\vec{k})}{z^{-1} e^{\beta \epsilon(\vec{k})} - \eta}$$

$$\beta P_\eta = \frac{g}{\lambda^3} f_{5/2}^\eta(z)$$

$$n_\eta = \frac{g}{\lambda^3} f_{3/2}^\eta(z)$$

$$\epsilon_\eta = \frac{3}{2} P_\eta$$

$$f_m^\eta(z) = \frac{1}{(m-1)!} \int_0^\infty \frac{dx}{x^{m-1}} z^{-1} e^x - \eta = \sum_{\alpha=1}^\infty \eta^{\alpha+1} \frac{z^\alpha}{\alpha^m}$$

$$\frac{d}{dz} f_m^\eta(z) = \frac{1}{z} f_{m-1}^\eta(z)$$

Problem 1 (35 points)

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Consider a polymer formed by connecting N disc-shaped molecules into a one-dimensional chain. Each molecule can align along either its long axis (of length $2a$) or short axis (of length a). Furthermore, when laying on its long axis there are two configuration it may assume, whereas when lying along its short axis there is only one possible configuration. The energy of the monomer aligned along its shorter axis is higher by ϵ , that is, the total energy of the system is $H = \epsilon U$ where U is the number of monomers standing up.

a (10 points) Calculate the partition function $Z(T, N)$ of the polymer.

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$$Z_N = \frac{(Z_1)^N}{N!} \checkmark$$

$$Z_1 = e^{-\beta \epsilon} + 1 \Rightarrow Z_N = \frac{1}{N!} (1 + 2e^{-\beta \epsilon})^N = Z(T, N)$$

laying long axis $U=1$ $\checkmark$ lying short axis $U=0$

with 2 configuration, standing up $\checkmark$

$$e^{-\beta H} = e^{-\beta \epsilon} \quad e^{-\beta H} = 1$$

$$Z_1 = \sum e^{-\beta H} = 2 \cdot e^{-\beta \epsilon} + 1$$

b (15 points) Calculate the average length $\langle L(T, N) \rangle$ and the variance of the length $\langle L(T, N)^2 \rangle_c$ of the polymer.

$$\langle L(T, N) \rangle = N \langle L_1(T, N) \rangle = \left(a \cdot \frac{1}{1+2e^{-\beta \epsilon}} + 2a \cdot \frac{2e^{-\beta \epsilon}}{1+2e^{-\beta \epsilon}} \right) \cdot N \checkmark$$

$$= N \frac{a}{1+2e^{-\beta \epsilon}} (1+4e^{-\beta \epsilon})$$

$$p = \frac{e^{-\beta H}}{Z}$$

$$\langle L(T, N)^2 \rangle_c = \langle L(T, N)^2 \rangle - \langle L(T, N) \rangle^2 = \left(a^2 \cdot \frac{1}{1+2e^{-\beta \epsilon}} + 4a^2 \frac{2e^{-\beta \epsilon}}{1+2e^{-\beta \epsilon}} \right) \cdot N - \left(\frac{Na}{1+2e^{-\beta \epsilon}} \right)^2 (1+4e^{-\beta \epsilon})^2$$

$$= \frac{Na^2}{1+2e^{-\beta \epsilon}} (1+8e^{-\beta \epsilon}) - \left(\frac{Na}{1+2e^{-\beta \epsilon}} \right)^2 (1+4e^{-\beta \epsilon})^2 \checkmark$$

c (10 points) Determine the energy per monomer and heat capacity per monomer. Sketch the heat capacity as a function of temperature.

$$\begin{aligned}
 \langle E \rangle &= - \frac{\partial \ln Z_1}{\partial \beta} = - \frac{\partial}{\partial \beta} \ln(1 + 2e^{-\beta \epsilon}) \\
 &= - \frac{1}{1 + 2e^{-\beta \epsilon}} - 2(-\epsilon)e^{-\beta \epsilon} = \frac{2\epsilon e^{-\beta \epsilon}}{1 + 2e^{-\beta \epsilon}} = \left(1 - \frac{1}{1 + 2e^{-\beta \epsilon}}\right) \cdot \epsilon
 \end{aligned}$$

$$C = \frac{\partial \langle E \rangle}{\partial T} = \frac{\partial \beta}{\partial T} \frac{\partial \langle E \rangle}{\partial \beta}$$

$$\beta = \frac{1}{k_B T}$$

$$\begin{aligned}
 &= \frac{-1}{k_B T^2} \frac{\epsilon \cdot 2 \cdot (-\epsilon) e^{-\beta \epsilon}}{(1 + 2e^{-\beta \epsilon})^2} = \frac{2\epsilon^2 e^{-\beta \epsilon}}{k_B T^2 (1 + 2e^{-\beta \epsilon})^2} \quad e^{-\beta \epsilon} = 1
 \end{aligned}$$

$$\text{For } T \rightarrow \infty \quad \beta \rightarrow 0 \quad e^{-\beta \epsilon} \rightarrow 1$$

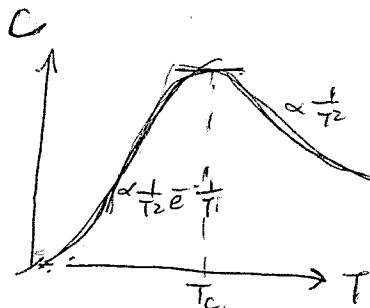
$$C \rightarrow \frac{2\epsilon^2}{k_B T^2 (1+2)^2} = \frac{2\epsilon^2}{9k_B T^2} \propto \frac{1}{T^2}$$

$$\text{For } T \rightarrow 0 \quad \beta \rightarrow \infty \quad e^{-\beta \epsilon} \rightarrow 0$$

$$C \rightarrow \frac{2\epsilon^2 e^{-\beta \epsilon}}{k_B T^2} \propto \frac{1}{T^2} e^{-\frac{\epsilon}{T}}$$

dominant at low T

$$x \rightarrow 0$$



Critical Temperature

$$\frac{\partial C}{\partial T} = \frac{\partial C}{\partial \beta} \frac{\partial \beta}{\partial T} = 0$$

$$\begin{aligned}
 \Rightarrow \left. \frac{\partial C}{\partial \beta} \right|_{T=T_c} &= 0 = \frac{4\epsilon^2 k_B}{T_c^2} \frac{e^{-\beta \epsilon}}{(1 + 2e^{-\beta \epsilon})^2} \\
 &= 4\epsilon^2 k_B \left(2\beta \frac{e^{-\beta \epsilon}}{(1 + 2e^{-\beta \epsilon})^2} + \beta^2 \right)
 \end{aligned}$$

Problem 2 (35 points)

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The pressure P of a gas is related to its density $n = N/V$ and temperature T by the truncated expansion:

$$P = k_B T n - \frac{b}{2} n^2 + \frac{c}{6} n^3$$

where b and c are assumed to be positive temperature-independent constants.

- a (15 points) Locate the critical temperature T_c below which the equation must be invalid, and the corresponding density n_c and pressure P_c of the critical point. Find the ratio $k_B T_c n_c / P_c$

Under critical value. T_c, n_c, P_c , $\frac{\partial P}{\partial n} = \frac{\partial^2 P}{\partial n^2} = 0$ ✓

$$\left. \frac{\partial P}{\partial n} \right|_{\substack{n=n_c \\ T=T_c}} = 0 = k_B T - b n + \frac{c}{2} n^2 \Rightarrow T_c = \frac{1}{k_B} (b n_c - \frac{c}{2} n_c^2)$$

$$\left. \frac{\partial^2 P}{\partial n^2} \right|_{n=n_c} = 0 = -b + c n_c \Rightarrow n_c = \frac{b}{c} \quad \checkmark \quad T_c = \frac{1}{k_B} \left(b \cdot \frac{b}{c} - \frac{c}{2} \left(\frac{b}{c} \right)^2 \right) = \frac{b^2}{2 c k_B} \checkmark$$

$$\Rightarrow P_c = P(T_c, n_c) = k_B \cdot \frac{b^2}{2 c k_B} \cdot \frac{b}{c} - \frac{b}{2} \left(\frac{b}{c} \right)^2 + \frac{c}{6} \left(\frac{b}{c} \right)^3 = \frac{b^3}{2 c^2} - \frac{b^3}{2 c^2} + \frac{b^3}{6 c^2} = \frac{b^3}{6 c^2} \checkmark$$

$$\text{Ratio } \frac{k_B T_c n_c}{P_c} = \frac{k_B \cdot \frac{b^2}{2 c k_B} \cdot \frac{b}{c}}{b^3 / 6 c^2} = \frac{6}{2} = 3 \quad \checkmark$$

b (10 points) Calculate the isothermal compressibility $\kappa_T = -\frac{1}{V} \frac{\partial V}{\partial P} \Big|_T$ and sketch its behavior as a function of T for $n = n_c$.

$$6 \quad \kappa_T = -\frac{1}{V} \frac{\partial V}{\partial P} \Big|_T = -\frac{1}{(N/n)} \frac{\partial (N/n)}{\partial P} \Big|_T = -n \frac{\partial (1/n)}{\partial P} \Big|_T$$

$$\frac{\partial P}{\partial (1/n)} = \frac{\partial P}{\partial n} \left(\frac{\partial (1/n)}{\partial n} \right) = -\frac{1}{n^2} \frac{\partial P}{\partial n} \Rightarrow \kappa_T = -n \left(-\frac{1}{n^2} \frac{\partial P}{\partial n} \right)^{-1} = n^3 \left(\frac{\partial P}{\partial n} \right)^{-1}$$

$$\frac{\partial P}{\partial n} = k_B T - bn + \frac{C}{2} n^2 \quad \kappa_T = n^3 \left(k_B T - bn + \frac{C}{2} n^2 \right)^{-1}$$

$$\kappa_T(T) \Big|_{n=n_c} = n_c^3 \left(k_B T - bn_c + \frac{C}{2} n_c^2 \right)^{-1}$$

$$n_c = \frac{b}{C}$$

$$= \left(\frac{b}{C} \right)^3 \left(k_B T - \frac{b^2}{C} + \frac{b^2}{2C} \right)^{-1} = \left(\frac{b}{C} \right)^3 \left(k_B T - \frac{b^2}{2C} \right)^{-1}$$

$$P = k_B T n - \frac{b}{2} n^2 + \frac{C}{6} n^3$$

$$\frac{k_B T_c n_c}{C} = \frac{b^2}{2C}$$

c (10 points) On the critical isotherm give an expression for $(P - P_c)$ as a function of $(n - n_c)$

$$1 \cup \quad P - P_c = k_B T_c n - \frac{b}{2} n^2 + \frac{C}{6} n^3 - \frac{b^3}{6C^2}$$

$$(n - n_c)^3$$

$$n_c = \frac{b}{C}$$

$$= \frac{C}{6} \left(-n_c^3 + n^3 - 3 \left(\frac{b}{C} \right)^2 n + 3 \left(\frac{b}{C} \right) n^2 \right)$$

$$\frac{C}{6} \quad \frac{b}{C}$$

$$= \frac{C}{6} (n - n_c)^3 \quad \checkmark$$

change $\frac{b}{C}$ into n_c .

$$T_c = \frac{b^2}{2Ck_B}$$

Problem 3 (30 points + 10 bonus)

Molecular hydrogen exists in two states: ortho and para.

- a (20 points) In para-hydrogen, the two electrons of H_2 form a singlet (antisymmetric) state. Therefore, the orbital angular momentum can have only even values. That is, the Hamiltonian has the form:

$$H = \frac{\hbar^2}{2I} l(l+1) \quad \text{para}$$

for $l = 0, 2, 4, \dots$

In contrast, in ortho-hydrogen, the electrons are in a triply degenerate symmetric state. Therefore, the orbital angular momentum can have only odd values. That is, the Hamiltonian has the form:

$$H = \frac{\hbar^2}{2I} l(l+1) \quad \text{ortho}$$

for $l = 1, 3, 5, \dots$

Compute the rotational partition function for both para-hydrogen and ortho-hydrogen, and evaluate both of their low and high temperature limits.

$$Z_{\text{para}} = \sum_{l, \text{even}} e^{-\beta \frac{\hbar^2}{2I} l(l+1)} = \sum_{n=0}^{\infty} e^{-\beta \frac{\hbar^2}{2I} 2n(2n+1)}$$

$$\Rightarrow l = 2n$$

$$n \in \mathbb{N} + \{0\}$$

$$Z_{\text{ortho}} = \sum_{l, \text{odd}} e^{-\beta \frac{\hbar^2}{2I} l(l+1)} = \sum_{n=0}^{\infty} e^{-\beta \frac{\hbar^2}{2I} (2n+1)(2n+2)} \cdot 3$$

$l = 2n+1$
 $n \in \mathbb{N} + \{0\}$

$\rightarrow$ triply degenerate.

At low T $\beta \rightarrow \infty$ $e^{-\beta} \rightarrow 0$, take the first few terms.

$$\rightarrow Z_{\text{para}} \approx 1 + e^{-\beta \frac{\hbar^2}{2I} \cdot 2 \cdot 3} = 1 + 5e^{-3\beta \frac{\hbar^2}{I}}$$

$$Z_{\text{ortho}} \approx 3 \cdot e^{-\beta \frac{\hbar^2}{2I} \cdot 1 \cdot 2} = 3e^{-\beta \frac{\hbar^2}{I}} \cdot 3 = 9e^{-\beta \frac{\hbar^2}{2I}}$$

At high T , $(2n+1)(2n+2) \rightarrow 4n^2$ $2n(2n+1) \rightarrow 4n^2$ $4n+1 \rightarrow 4n$ $4n+3 \rightarrow 4n$

$$Z_{\text{para}} \approx \int_0^{\infty} (e^{-\beta \frac{\hbar^2}{2I} \cdot 4n^2})_{4n dn} = \int_0^{\infty} e^{-\frac{2\beta \hbar^2}{I} n^2} \frac{4n dn}{\left(\frac{2 dn^2}{2 dn^2}\right)} = 2 \frac{I}{\beta \hbar^2} \int_0^{\infty} e^{-x} dx = \frac{I}{\beta \hbar^2}$$

$x = \frac{2\beta \hbar^2}{I} n^2$

$$Z_{\text{ortho}} \approx \int_0^{\infty} 3 \cdot e^{-\beta \frac{\hbar^2}{2I} \cdot 4n^2} \frac{4n dn}{4n dn} = 3 Z_{\text{para}} = 3 \cdot \frac{I}{\beta \hbar^2}$$

- b (10 points) For an equilibrium gas of N hydrogen molecules composed of N_p para and $N_o = N - N_p$ ortho molecules, calculate the partition function of the complete system. (you may ignore vibrational degrees of freedom)

$$Z = \sum_{N_o, N_p} \frac{Z_o^{N_o} Z_p^{N_p}}{N_o! N_p!} = \sum_{N_p} \frac{Z_o^{N-N_p} Z_p^{N_p}}{(N-N_p)! N_p!} = \frac{1}{N!} \sum_{N_p} \frac{N!}{(N-N_p)! N_p!} Z_o^{N-N_p} Z_p^{N_p}$$

Binomial

$$= \frac{1}{N!} (Z_o + Z_p)^N$$

where Z_o is the ortho hydrogen rotational partition function
 Z_p is the para hydrogen rotational partition function listed in part a.

- c (10 points) Compute an expression for the rotational contribution to the internal energy $\langle E_{rot} \rangle$. Determine its behavior at the high and low temperature limits. Note that due to the large transition times between ortho- and para-hydrogen, in most circumstances the mixture is not actually in equilibrium.

$$\langle E_{rot} \rangle = - \frac{\partial \ln Z}{\partial \beta} = - \frac{\partial}{\partial \beta} (N \ln(Z_o + Z_p) - \ln N!) = -N \frac{\partial}{\partial \beta} \ln(Z_o + Z_p)$$

For Low temp, $e^{-x} \rightarrow 0$ $x = \frac{3\beta \hbar^2}{2I}$, neglect e^{-2x} term

$$\ln(1+x) \approx$$

$$\ln(Z_o + Z_p) = \ln(1 + 5e^{-\frac{3\beta \hbar^2}{2I}} + 9e^{-\frac{3\beta \hbar^2}{2I}}) \approx \ln(1 + 9e^{-\frac{3\beta \hbar^2}{2I}})$$

$$\frac{\partial}{\partial \beta} \ln(Z_o + Z_p) = \frac{\partial \ln(1 + 9e^{-\frac{3\beta \hbar^2}{2I}})}{\partial (1 + 9e^{-\frac{3\beta \hbar^2}{2I}})} \frac{\partial (1 + 9e^{-\frac{3\beta \hbar^2}{2I}})}{\partial \beta} = \frac{1}{1 + 9e^{-\frac{3\beta \hbar^2}{2I}}} \cdot 9 \left(-\frac{3\hbar^2}{2I}\right) e^{-\frac{3\beta \hbar^2}{2I}}$$

$$\langle E_{rot} \rangle \approx \frac{\partial \ln(Z_o + Z_p)}{\partial \beta} = \frac{\partial \ln(1 + 9e^{-\frac{3\beta \hbar^2}{2I}})}{\partial \beta} = \frac{1}{1 + 9e^{-\frac{3\beta \hbar^2}{2I}}} \cdot 9 \left(-\frac{3\hbar^2}{2I}\right) e^{-\frac{3\beta \hbar^2}{2I}}$$

$$Z_{para} \approx \sqrt{\frac{\pi}{\beta}} \cdot \frac{\hbar}{2} \quad \frac{\partial}{\partial \beta} Z_{para} = -\frac{1}{2} \beta^{-1} Z_{para}$$

For High Temp

$$\ln(Z_o + Z_p) = \ln(4 Z_{para}) = \ln Z_{para} + \ln 4$$

$$\frac{\partial}{\partial \beta} \ln(Z_o + Z_p) = \frac{\partial}{\partial \beta} \ln(Z_{para}) = \frac{1}{Z_{para}} \frac{\partial}{\partial \beta} Z_{para} = -\frac{1}{2} \beta^{-1} = -\frac{1}{2} k_B T$$

$$\langle E_{rot} \rangle \approx - \frac{\partial \ln Z}{\partial \beta} = \frac{1}{2} k_B T$$

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Have a good break!